



Praktikum Design of Wind Farms

Dr. sc. Hadi Hoghooghi – hadi.hoghooghi@tum.de

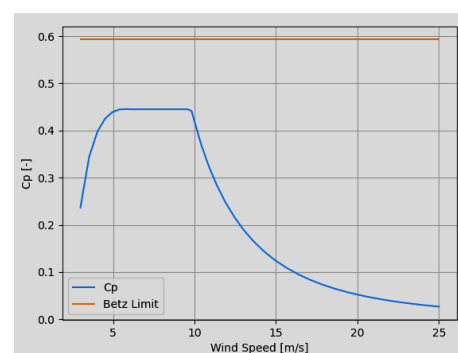
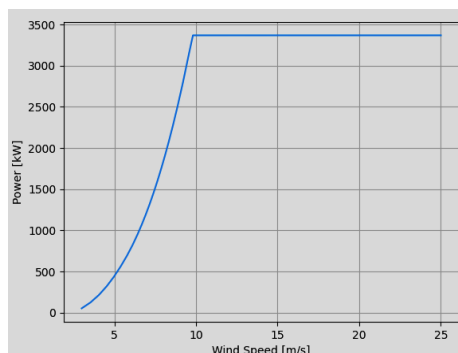
MSc. Samuel Kainz – samuel.kainz@tum.de

Assignment 5 (11%): Layout Optimization for LCoE

This assignment aims at introducing students to optimization problems in general and to cost-based layout optimization specifically, including understanding an open-source gradient-free optimization algorithm, setting up the objective function dependent on the design variables, defining constraints and boundaries, investigating convergence, and analysing the results delivered by the optimization algorithm in the context of wind farm layout optimization. Furthermore, students learn how to set up FLORIS based on wind resource time series retrieved from a wind atlas.

Eight IEA 3.4MW wind turbines are planned to be installed around Munich within a square boundary (2.8 x 2.8 km) and with a minimum separation distance of 2D between all turbines. The substation is placed in the center point of the boundary. The wind shear exponent and TI are 0.20 and 6%, respectively. The key parameters of the turbines and their characteristics are as follows:

Item	Value	Units
Name	IEA 3.4MW Reference	N/A
Rated Power	3370	kW
Rated Wind Speed	9.8	m/s
Cut-in Wind Speed	3	m/s
Cut-out Wind Speed	25	m/s
Rotor Diameter	130	m
Hub Height	120	m
Control	Pitch Regulated	N/A



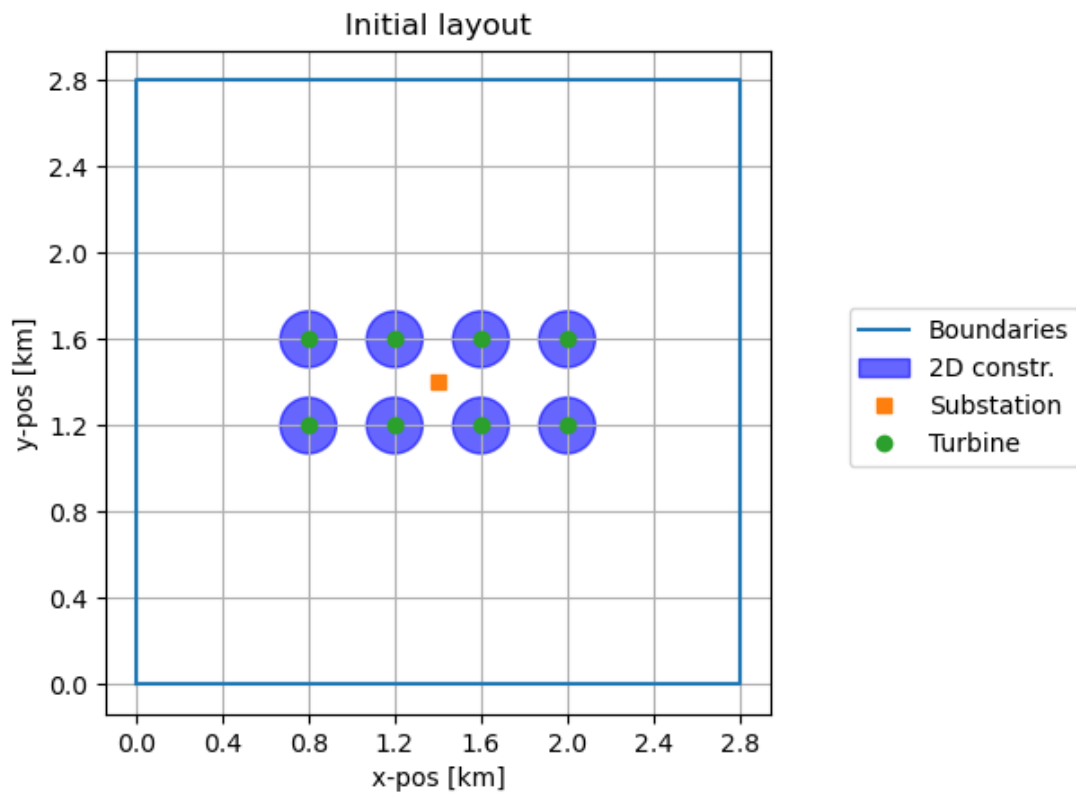
A .csv file is available on [GitHub](#).



The following financial and project parameters are to be considered:

Item	Value	Units
Nominal discount rate	4	%
Turbine costs	$1.3 \cdot 10^6$	\$/MW
O&M costs	0.016	\$/kWh
Rental costs	20,000	\$/MW-yr
Fuel cost	5.19	USD / gal
Electricity price	67	\$/MWh
Line frequency	50	Hz
Array voltage	30	kV
Project construction time	12	months
Project lifetime	20	years

The initial layout suggested by non-experts looks like the following:





1. Create and plot the wind rose at hub height in FLORIS based on a time series of wind speed and wind direction at *Marienplatz (Munich)* extracted from the [New European Wind Atlas](#) for the whole time history available on the website (01/01/2005 - 31/12/2018).
Hints: download the “raw”-data at 10m height and scale to hub height; *.nc files can be easily loaded using the netCDF4 package; to feed time-series into a windrose object, check the example “[Wind Data Objects](#)” for FLORIS v4, or the function “make_wind_rose_from_user_data” for FLORIS v3.
2. Evaluate AEP, capacity factor, and farm efficiency using the Gauss wake model (default parameters) of the baseline wind farm. Then, calculate LCoE and PI. Interpret the results based on the knowledge you gained in the previous assignments and lectures of the course.
3. Install the scipy package and familiarize with the “[minimize](#)” function in the optimization package. Use the gradient-free “Constrained Optimization BY Linear Approximation” (COBYLA) method to optimize the turbine positions in order to minimize the LCoE of the wind farm. Respect the boundaries and the constraint for minimum turbine distance. Make a plot containing the boundaries, substation position, and the initial and final layout. Furthermore, plot the convergence of the objective function over the number of iterations. Compare AEP, capacity factor, farm efficiency, LCoE, and PI of the optimized layout with the initial layout. Interpret the results and discuss the changes in turbine positions. Has the project become profitable? Hints:
 - Define one Python function that calculates the required parameters in dependence of the turbine positions (= input variable)
 - Use the plots mentioned above to make sure you found a (local) optimum and that the constraints have been met
 - Adjust the function options to ensure that the optimization converges

Note: Please upload your code and a Word/PDF file in which you describe the steps of your solution and show/explain your results. It is not enough to only upload code files.